PART 1 · MODULE A

Computer and internet fundamentals

Before you build anything for the web, you need a clear picture of what happens when someone opens a page.

A1. The computer in simple words

Simple explanation. A computer stores information, calculates, and shows results. An operating system (Windows, macOS, Android, iOS or Linux) manages the hardware so programs can use it.

Everyday analogy. A school: the storeroom keeps files (storage), teachers do the work (processor), the notice board shows results (screen). The principal who keeps everything running is the operating system.

A2. The internet and the browser

Simple explanation. The internet is a network of computers that send messages to each other. A browser (Chrome, Edge, Firefox, Safari) asks a computer for a web page and displays it.

Everyday analogy. Ordering food online: you pick a shop (website), the kitchen prepares the order (server), and a delivery brings it to your door (the internet).

A3. Websites, web apps, domains and hosting

Term	Simple meaning	Example
Website	Pages that show information	A bakery's menu
Web application	A website where you can do something (log in, save data, pay)	A booking form that saves appointments
Domain	The address people type, such as `example.com`	The shop's name on the street
Hosting	A computer that stays on and serves your site	The shop building you rent

A4. Servers, databases and client–server architecture

Simple explanation. The **client** (usually a browser) asks for something. The **server** answers. A **database** is an organised store, like a school register with one row per student.

TEXT

```
Client (browser) --- request: GET /products ---> Server
Client (browser) <--- response: product list ---- Server
                                                    |
                                                    v
                                        Database (products table)
```

A5. HTTP and HTTPS

HTTP is the language browsers and servers use. A request has a **method** (GET to read, POST to send data), an **address** and **headers**. The reply has a **status code**: 200 means success, 404 means not found, 500 means the server failed. HTTPS is HTTP with encryption. Every public site you build must use HTTPS.

A6. Basic cybersecurity for beginners

- Use a password manager and two-factor authentication on every work account.
- Never paste secret keys into a website, a chat or a public repository.
- Be sceptical of unexpected invoices, login pages and urgent payment requests.

ASSIGNMENT

Map a page request

Business problem: A client wants staff to understand what happens when a customer opens the shop's website. Draw the flow.

Steps

1. Write the steps from typing the address to seeing the page.
2. Label each step as client, server, network or database.
3. Mark the step where HTTPS protects the data.

Expected output: A numbered list of six to eight labelled steps.

SOLUTION (ANSWER KEY)

1. The customer types `shop.example.com` in the browser (client).
2. The browser asks the domain name system for the server's address (network).
3. The browser sends `GET /` over HTTPS (client to server, encrypted).
4. The server asks the database for the products (server to database).
5. The database returns the rows; the server builds an HTML page and replies with status 200 (server).
6. The browser displays the page (client).

Why it works: the steps follow the client–server model. Step 3 is where encryption protects the request on its way.

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